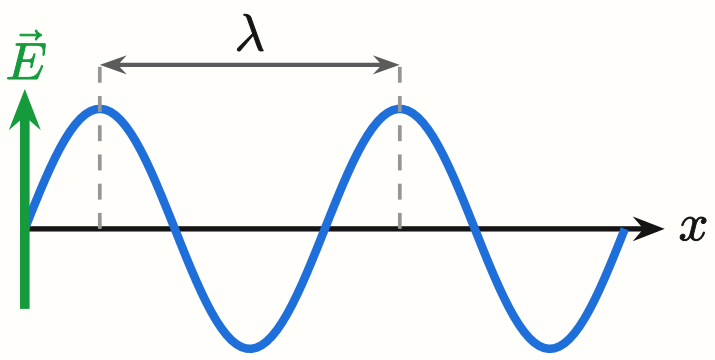
Wave Optics

Class 12 · MONOCHROMATIC light · λ in VACUUM (air) unless stated · COHERENT sources

1 Light is a transverse wave



- $\vec{E}$ oscillates PERPENDICULAR to the direction of travel (transverse); only transverse waves can be polarised.
- Speed in a medium $v = c/\mu$; the FREQUENCY ν is fixed by the source, so $\lambda_{medium} = \lambda/\mu$.

2 Path difference and phase difference

$$\phi = \frac{2\pi}{\lambda}\Delta x \qquad \Delta x = \mu L \text{ (optical path)}$$

Δx = path difference (m), ϕ = phase difference (rad), λ = wavelength in VACUUM · geometric path L in a medium of refractive index μ counts as μL · $\Delta x = \lambda \Leftrightarrow \phi = 2\pi$

- SIGN CONVENTION on every page: $\Delta x = (\text{path from } S_2) - (\text{path from } S_1)$, $\phi = \frac{2\pi}{\lambda}\Delta x$; y on the screen is measured from the centre O, positive on the side of S_1.
- Intensity $I \propto A^2$ (square of the amplitude); a detector records the TIME-AVERAGED intensity.

Syllabus Checklist

JEE Main + JEE Advanced + NEET syllabus + standard results

1 JEE Main + JEE Advanced + NEET syllabus

- ✓ Wavefronts; Huygens' principle — p3, p4
- ✓ Reflection and refraction by Huygens' principle — p5–p7
- ✓ Coherent sources; superposition, interference — p8
- ✓ $I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$, I_{max}, I_{min}, visibility — p9, p10
- ✓ YDSE: $\beta = \lambda D/d$, bright and dark fringes — p11, p12
- ✓ Angular fringe width; intensity on the screen — p12, p13
- ✓ YDSE in a medium (β/μ); white-light fringes — p14
- ✓ Thin sheet: shift $(\mu - 1)tD/d$ — p15
- ✓ Source slit displaced; oblique incidence — p16
- ✓ Thin films (reflected, transmitted); Lloyd's mirror — p17–p19
- ✓ Single slit: $a\sin\theta = n\lambda$, width $2\lambda D/a$, graph — p20, p21
- ✓ Fresnel distance; diffraction vs interference — p22
- ✓ Resolving power: microscope, telescope (Rayleigh) — p23
- ✓ Polaroids, Malus's law — p24, p25
- ✓ Brewster's law; polarisation by scattering — p26, p27
- ✓ Doppler effect for light ($v \ll c$) — p28

2 Also covered

- ✓ Special cases; standard results; common traps — p29–p31
- ✓ Solved questions (8 Main + 6 Advanced level) — p32–p36

Wavefronts

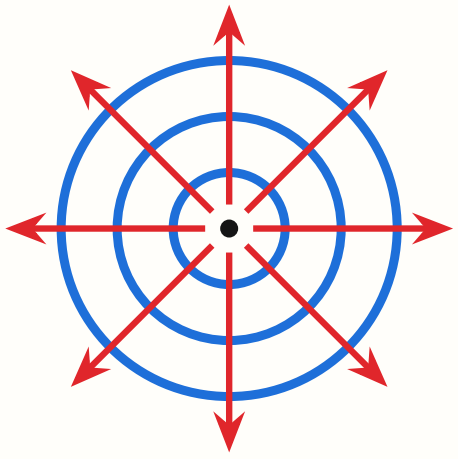
homogeneous ISOTROPIC medium (same speed in every direction)

1 What is a wavefront?

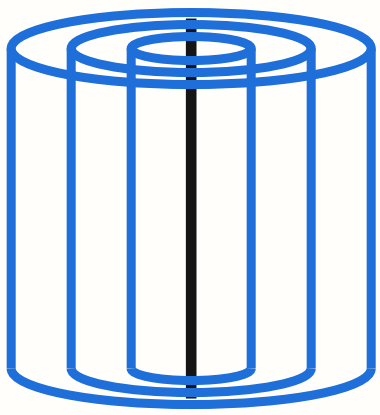
wavefront = surface of CONSTANT PHASE

all points on it oscillate in step · the energy flows along RAYS, which are NORMAL to the wavefronts (isotropic medium) · successive wavefronts one λ apart travel at speed v

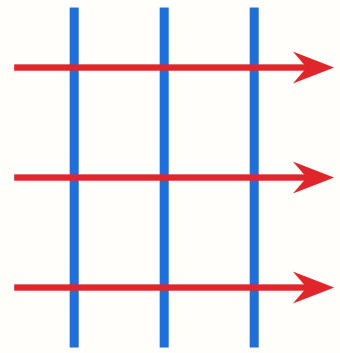
2 Three shapes



point source: SPHERICAL



line source: CYLINDRICAL



distant source: PLANE

- Point source: amplitude $\propto \frac{1}{r}$, intensity $\propto \frac{1}{r^2}$. Line source: amplitude $\propto \frac{1}{\sqrt{r}}$, intensity $\propto \frac{1}{r}$. Plane wave: amplitude constant (no absorption).
- A small part of a very large sphere is nearly flat: light from the sun or a star reaches us as a PLANE wavefront.

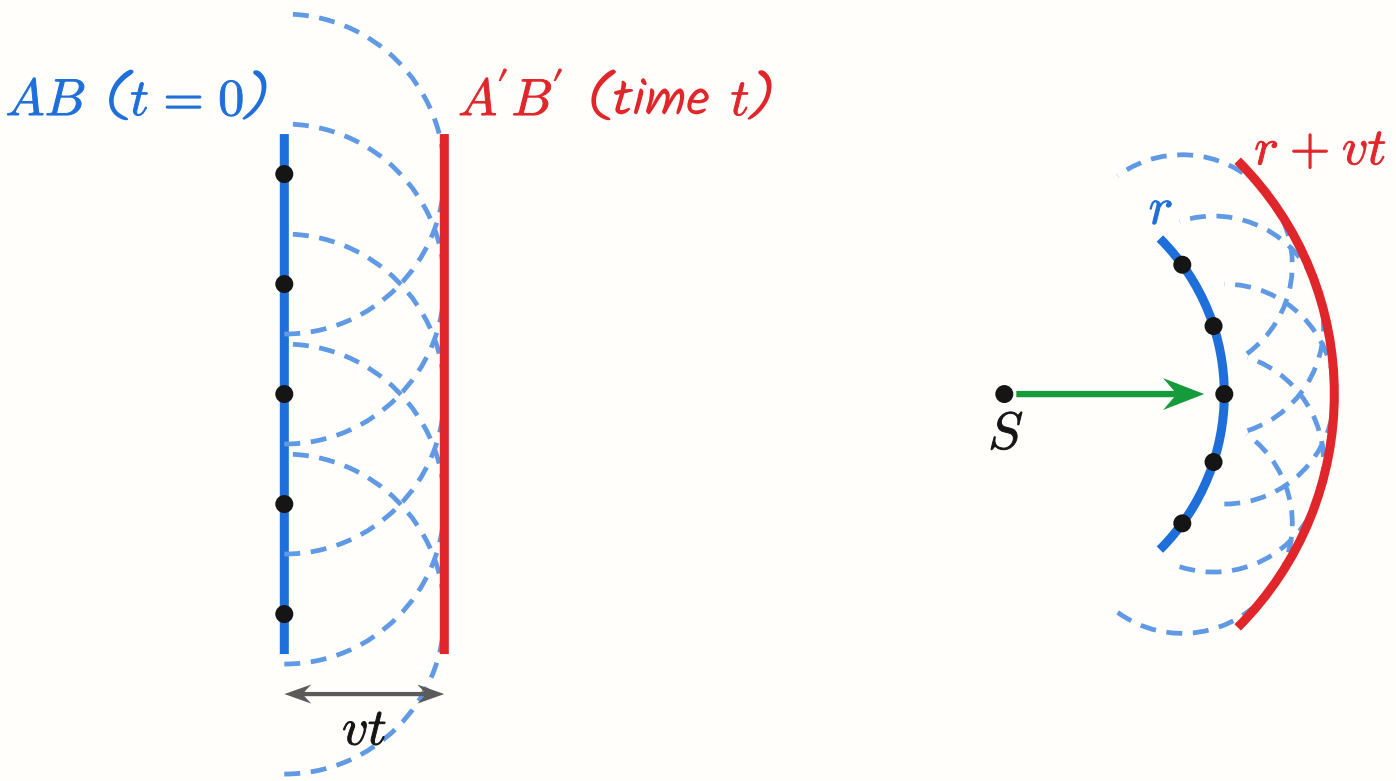
Huygens' Principle

homogeneous isotropic medium · wave speed v · forward wavelets only

1 Construction

- Every point of a wavefront acts as a source of SECONDARY WAVELETS, spreading with the wave speed v.
- After time t the new wavefront is the forward ENVELOPE (common tangent) of all wavelets of radius vt.
- The backward wave is ignored (Huygens assumed its amplitude is zero; the full wave theory confirms no backward wave).
- The new wavefront is perpendicular to the rays; Huygens' construction explains reflection and refraction (next pages).

2 Plane and spherical wavefronts

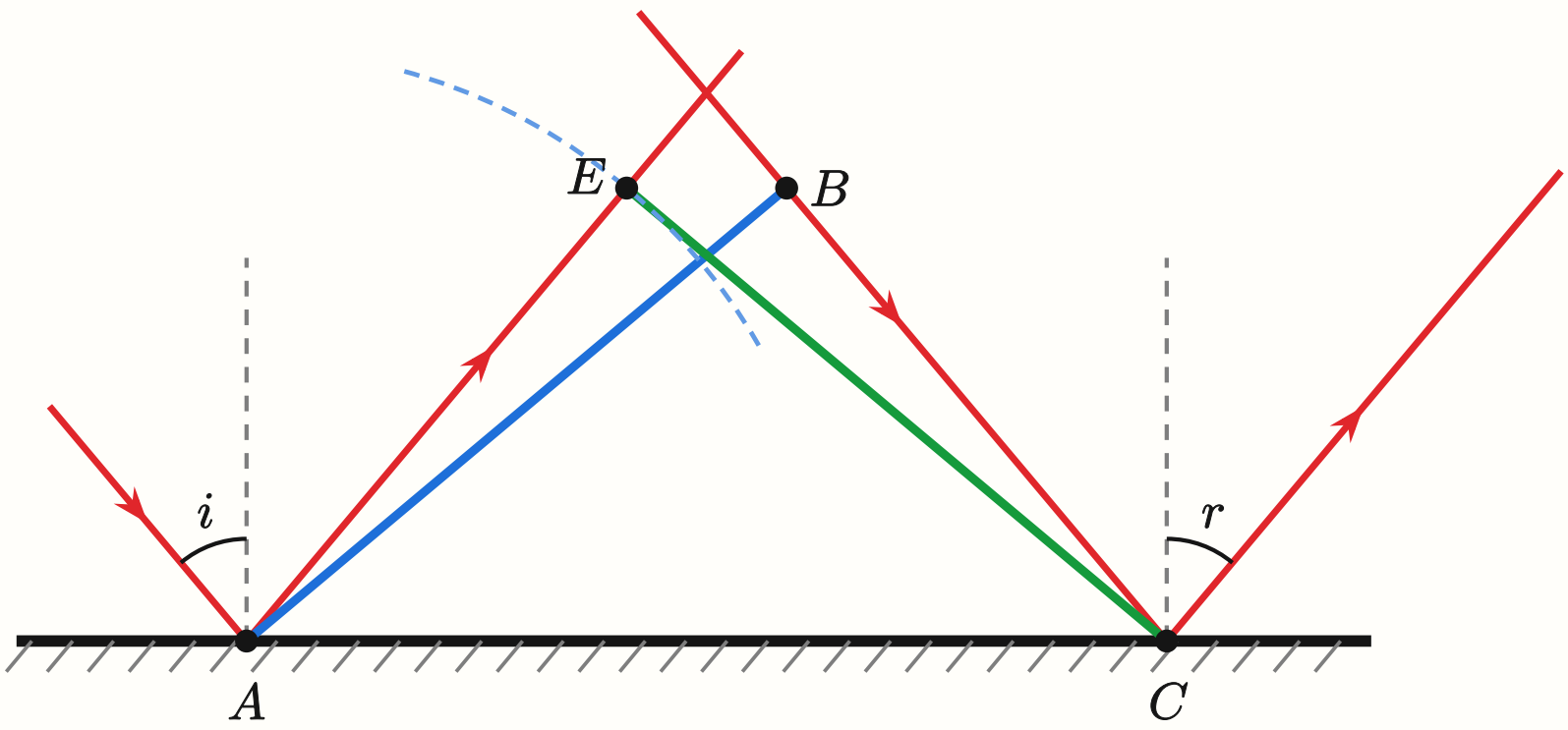


Dashed: secondary wavelets of radius vt. Red: the new wavefront (envelope), still plane / still spherical.

Reflection by Huygens' Principle

plane wavefront · plane mirror · same medium, speed v

1 Construction



Blue: incident wavefront AB; green: reflected wavefront EC. While B travels to C ($BC = vt$), the wavelet from A grows to radius $AE = vt$ (dashed arc).

2 Proof of the law of reflection

$$\triangle ABC \cong \triangle CEA \;\Rightarrow\; i = r$$

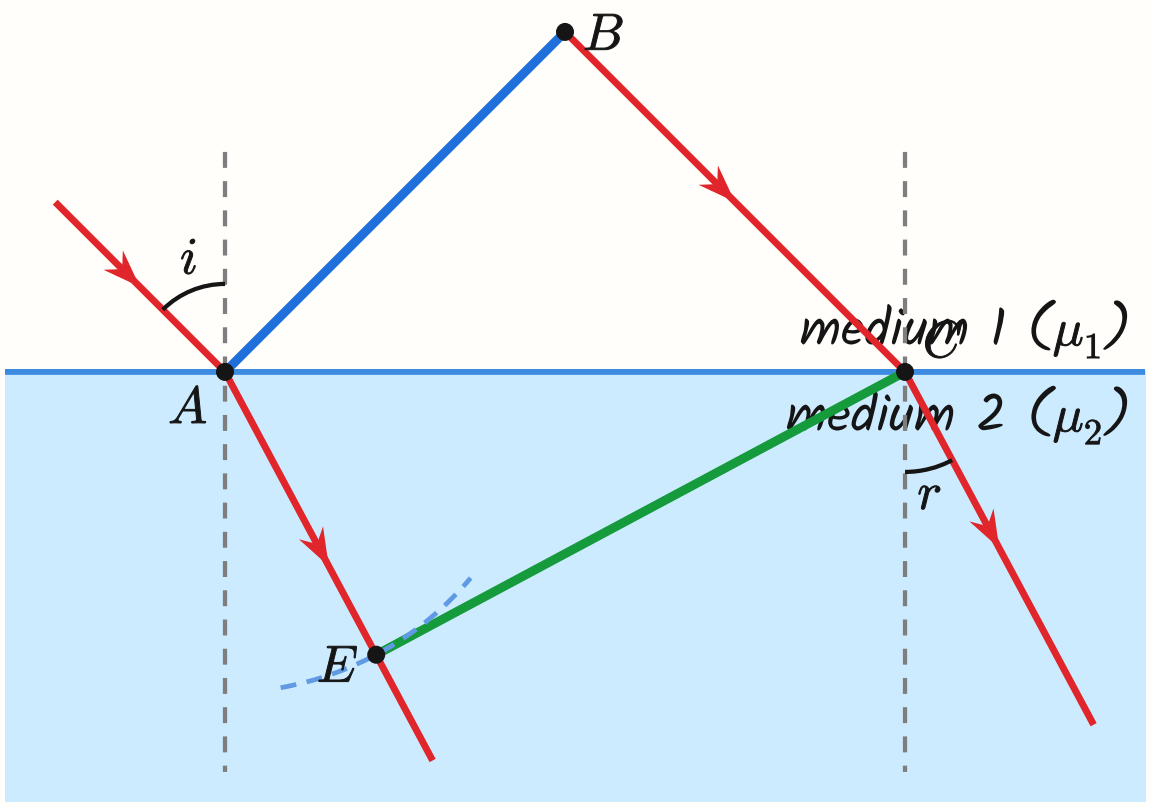
common side AC, $BC = AE = vt$, right angles at B and E (wavefront $\perp$ ray) · $\sin i = \frac{BC}{AC}$, $\sin r = \frac{AE}{AC}$ · incident ray, normal and reflected ray in ONE plane

- *The speed (and so λ) does not change on reflection: same medium.*
- *The reflected wavefront EC makes the same angle with the mirror as AB, turned the other way.*

Refraction by Huygens' Principle

plane wavefront · plane boundary · speeds v_1, v_2 in media 1, 2

1 Construction (here $v_2 < v_1$)



$BC = v_1 t$, $AE = v_2 t$: the wavefront slows down in medium 2 and turns. Green: refracted wavefront EC.

2 Snell's law

$$\frac{\sin i}{\sin r} = \frac{BC/AC}{AE/AC} = \frac{v_1}{v_2} = \frac{\mu_2}{\mu_1}$$

$\mu = \frac{c}{v}$ · i, r = angles with the normal · $v_2 < v_1$ (denser medium 2): $r < i$, bends TOWARDS the normal

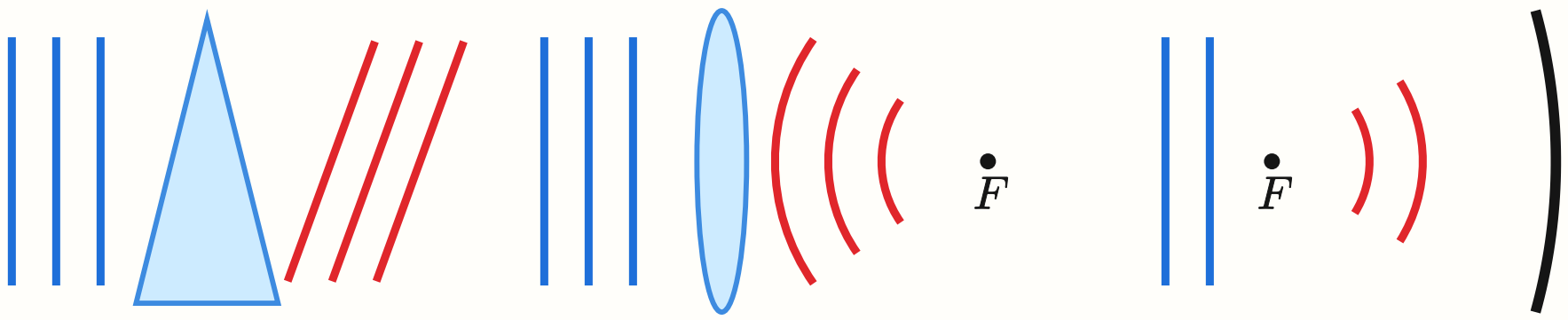
Example

$\lambda = 600\,\text{nm}$ (air) into water ($\mu = \frac{4}{3}$): $\nu = 5 \times 10^{14}\,\text{Hz}$ (unchanged), $v = 2.25 \times 10^{8}\,\text{m s}^{-1}$, $\lambda_w = 450\,\text{nm}$.

Wavefronts Through Optics

plane wavefront from the LEFT · thin prism, thin convex lens, concave mirror · paraxial

1 Prism, convex lens, concave mirror



prism: tilted — lens: converging — mirror: converging

Blue: incident plane wavefronts; red: emerging wavefronts. The red arcs are centred on the focus F and shrink towards it.

2 Why?

- Prism: the ray near the BASE crosses more glass, so that part of the wavefront is delayed: it tilts and the light deviates towards the base.
- Convex lens: thickest at the centre, the centre is delayed most, the wavefront becomes a sphere converging to F.
- Concave mirror: the edges reach the mirror first and turn back first, giving a spherical wavefront converging to F.

Golden points

★ Optical path μL is the same along EVERY ray from one wavefront to the next: this is why a lens forms an image (equal optical paths to the focus).

Superposition and Coherence

waves of the SAME frequency · linear medium · same polarisation for full interference

1 Principle of superposition

$$\vec{E} = \vec{E}_1 + \vec{E}_2$$

the resultant displacement (field) at a point = VECTOR sum of the displacements of the separate waves · each wave travels on unaffected

2 Coherent sources

- COHERENT: same frequency and a CONSTANT phase difference in time; the interference pattern stays fixed and can be seen.
- Two independent lamps (or two independent lasers) are INCOHERENT: each emits in short random bursts, the phase difference changes randomly and the pattern averages out: $I = I_1 + I_2$.
- Energy is only REDISTRIBUTED: the average intensity over a fringe is $I_1 + I_2$.

3 Two ways to get coherent sources

- Division of WAVEFRONT: two parts of one wavefront: Young's double slit, Lloyd's mirror, Fresnel biprism.
- Division of AMPLITUDE: partial reflection and transmission of the same wave: thin films, soap bubbles, oil on water.

- Waves polarised at right angles do not interfere (their fields cannot cancel).

Interference: Intensity

two COHERENT waves, same frequency and polarisation · ϕ = phase difference at the point

1 Resultant intensity

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$$

$$A^2 = A_1^2 + A_2^2 + 2A_1 A_2 \cos\phi$$

I_1, I_2 = intensities of the two waves alone · A = amplitude, $I \propto A^2$ · INCOHERENT sources: the $\cos\phi$ term averages to zero

$$I_{max} = \left(\sqrt{I_1} + \sqrt{I_2}\right)^2 \qquad I_{min} = \left(\sqrt{I_1} - \sqrt{I_2}\right)^2$$

constructive: $\phi = 2n\pi$, $\Delta x = n\lambda$ · destructive: $\phi = (2n-1)\pi$, $\Delta x = (2n-1)\frac{\lambda}{2}$ · $n = 0, \pm1, \pm2, \ldots$ (bright), $n = 1, 2, \ldots$ (dark, each side)

2 Visibility (contrast)

$$V = \frac{I_{max} - I_{min}}{I_{max} + I_{min}} = \frac{2\sqrt{I_1 I_2}}{I_1 + I_2}$$

$0 \le V \le 1$ · $V = 1$ (perfect dark fringes) only when $I_1 = I_2$

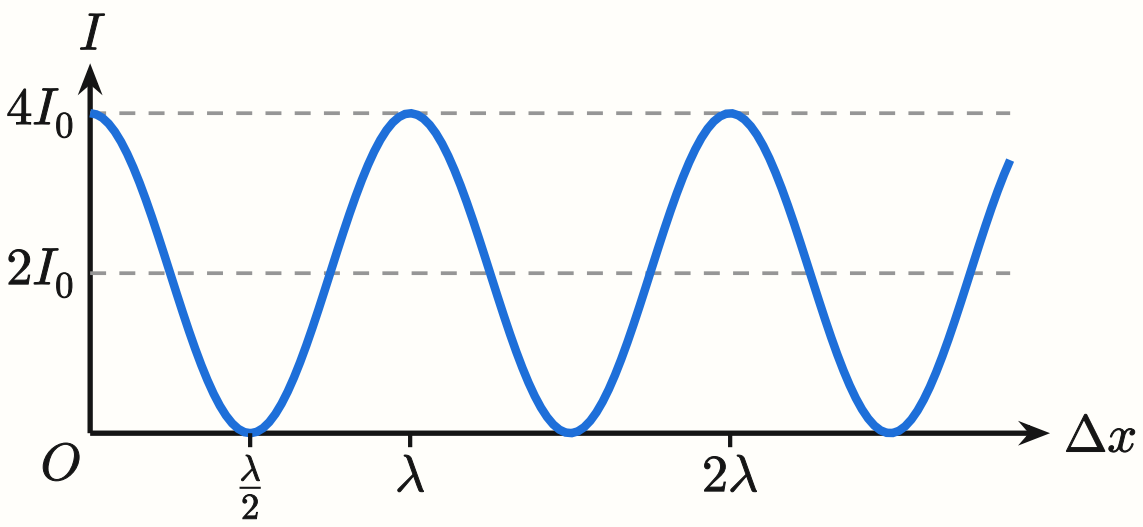
Example

$I_1 : I_2 = 4 : 1$: amplitudes $2 : 1$, $\frac{I_{max}}{I_{min}} = \frac{(2+1)^2}{(2-1)^2} = 9$, $V = \frac{2\times2}{4+1} = 0.8$.

Intensity Graphs

two COHERENT waves · Δx = path difference · $\phi = \frac{2\pi}{\lambda}\Delta x$

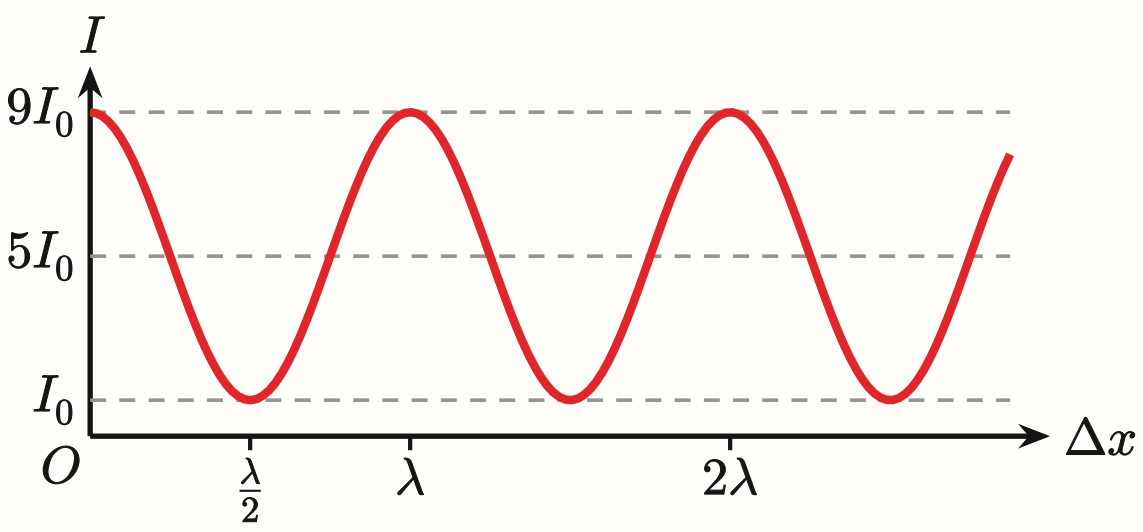
1 Equal intensities: $I_1 = I_2 = I_0$



$$I = 4I_0 \cos^2 \frac{\phi}{2}$$

max $4I_0$ at $\Delta x = n\lambda$ · ZERO at $\Delta x = \frac{\lambda}{2}, \frac{3\lambda}{2}$ · average $2I_0$ (dashed) · the curve starts at its MAXIMUM ($\Delta x = 0$)

2 Unequal: $I_1 = 4I_0$, $I_2 = I_0$



$$I = 5I_0 + 4I_0 \cos\phi$$

$I_{max} = 9I_0$, $I_{min} = I_0$ (never zero) · average $5I_0 = I_1 + I_2$ · $V = 0.8$

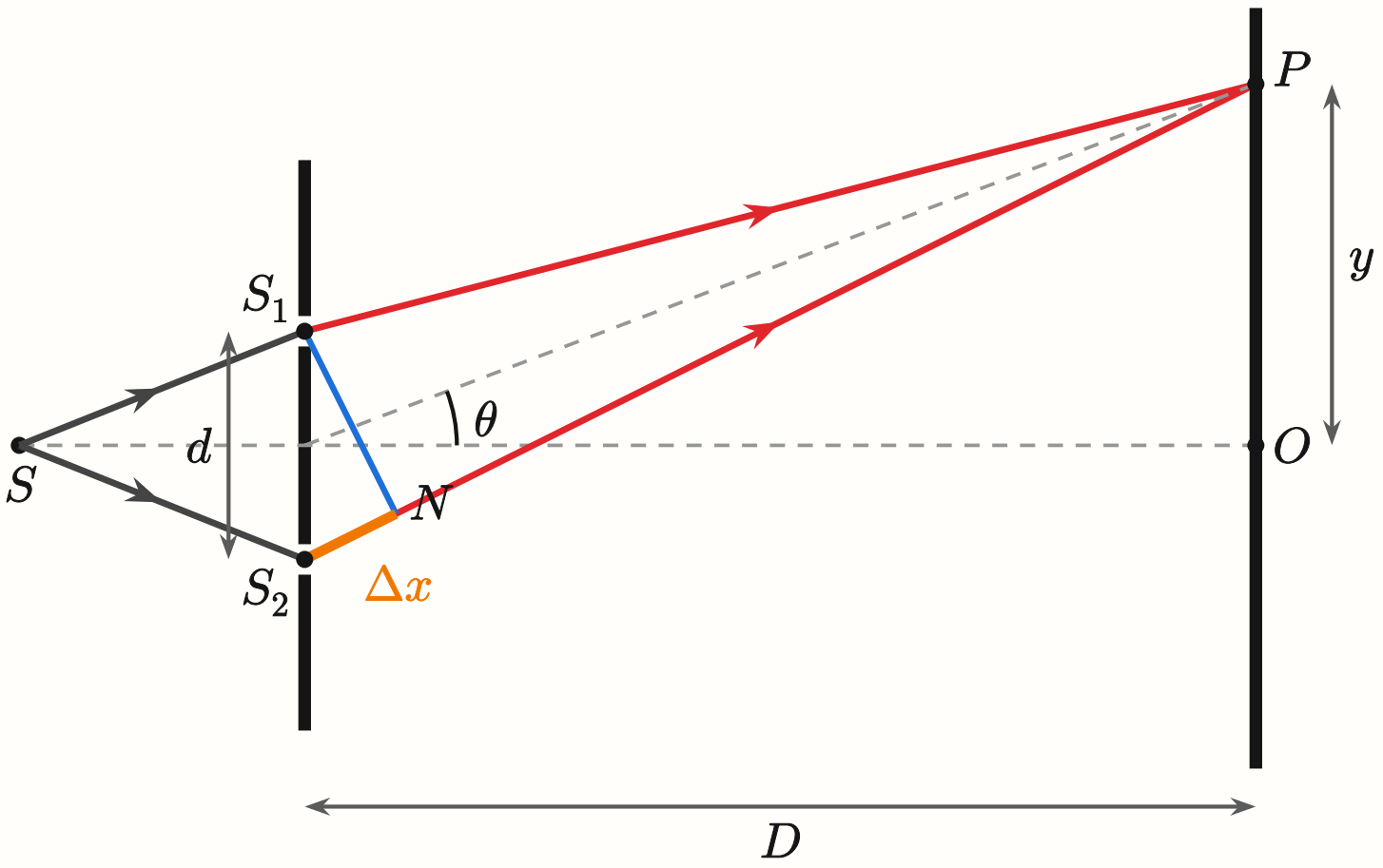
Golden points

★ Same positions of maxima and minima in both graphs: they depend on Δx only; the intensities decide the contrast.

YDSE: Geometry

Young's double slit · monochromatic, coherent slits S_1, S_2 (from one slit S) · in air

1 Set-up (not to scale)



$S_1N \perp S_2P$, so S_2N (orange) is the path difference. Real set-up: $D \approx 1\,\mathrm{m}$, $d < 1\,\mathrm{mm}$, so $S_1P \parallel S_2P$.

2 Path difference at P

$$\Delta x = S_2P - S_1P = d\sin\theta \approx d\tan\theta = \frac{yd}{D}$$

d = slit separation, D = slit-to-screen distance, y = distance of P from the centre O · $d\sin\theta$: $D \gg d$ (parallel rays) · $\approx \frac{yd}{D}$: SMALL θ as well

YDSE: Fringe Width

YDSE in air · monochromatic, coherent, equal slits · $D \gg d$ · SMALL angles ($y \ll D$)

1 Bright and dark fringes

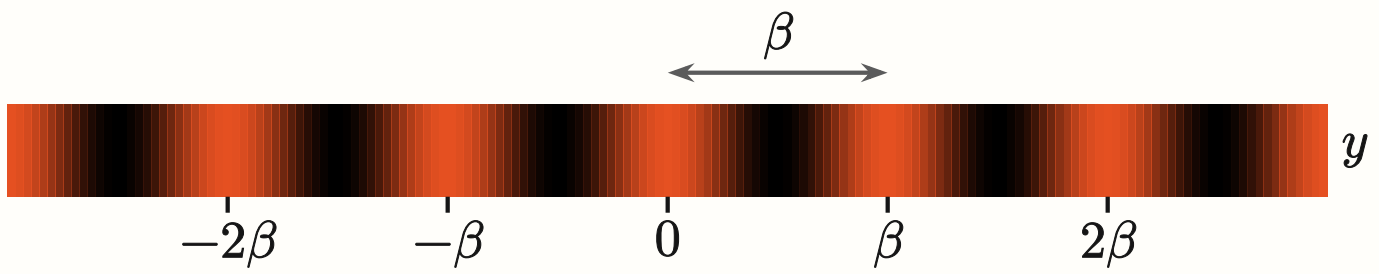
$$y_n = \frac{n\lambda D}{d} \text{ (bright)} \qquad y_n' = \frac{(2n-1)\lambda D}{2d} \text{ (dark)}$$

bright: $n = 0, \pm1, \pm2, \ldots$ ($n = 0$: central bright at O) · dark: $n = \pm1, \pm2, \ldots$ (sign = side) · n = order

$$\beta = \frac{\lambda D}{d} \qquad \theta_\beta = \frac{\beta}{D} = \frac{\lambda}{d}$$

β = fringe width = distance between two consecutive bright (or dark) fringes · θ_β = angular fringe width (rad) · bright and dark fringes have EQUAL widths

2 Pattern on the screen



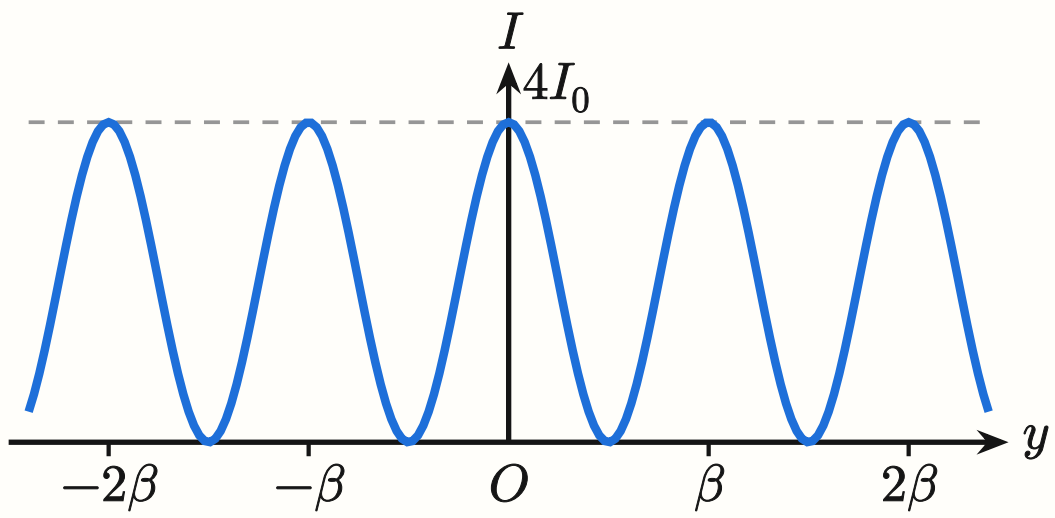
Example

$\lambda = 600\,\text{nm}$, $d = 0.50\,\text{mm}$, $D = 1.0\,\text{m}$: $\beta = \frac{600\times10^{-9}\times1.0}{0.50\times10^{-3}} = 1.2\,\text{mm}$;
3rd bright at $3.6\,\text{mm}$, 2nd dark at $1.8\,\text{mm}$ from O;
$\theta_\beta = 1.2 \times 10^{-3}\,\text{rad}$.

YDSE: Intensity and Fringe Count

YDSE · equal slits, each giving I_0 alone · coherent, monochromatic · $D \gg d$

1 Intensity on the screen



$$I = 4I_0 \cos^2 \frac{\pi y d}{\lambda D}$$

$= 4I_0 \cos^2 \frac{\pi y}{\beta}$ · small angles · maximum $4I_0$ at the centre $y = 0$ (equal paths)

2 How many maxima can be seen?

$$d \sin\theta = n\lambda, \ |\sin\theta| \le 1 \ \Rightarrow \ |n| \le \frac{d}{\lambda}$$

exact condition (any angle, $D \gg d$) · $\frac{d}{\lambda}$ NOT an integer: $2n_{max} + 1$ maxima, n_{max} = largest integer below $\frac{d}{\lambda}$ · screen catching all directions up to $90°$

Example

$d = 2.1\,\mu\text{m}$, $\lambda = 600\,\text{nm}$: $\frac{d}{\lambda} = 3.5$, so $n = 0, \pm1, \pm2, \pm3$: 7 maxima. Minima: $d\sin\theta = (2n-1)\frac{\lambda}{2}$ needs $|2n-1| < 7$: 6 minima ($\theta = 90°$ excluded).

Golden points

★ Far from the centre (large θ) the fringes are NOT equally spaced: $y = D\tan\theta$; $\beta = \frac{\lambda D}{d}$ holds only near the centre.

YDSE: Medium and White Light

YDSE · $D \gg d$ · small angles · λ = wavelength in vacuum (air)

1 Whole apparatus in a medium

$$\lambda' = \frac{\lambda}{\mu} \qquad \beta' = \frac{\lambda D}{\mu d} = \frac{\beta}{\mu}$$

the WHOLE space between slits and screen filled with a medium of refractive index μ · fringes come CLOSER

Example

$\beta = 1.2\,\text{mm}$ in air; the set-up is placed in water ($\mu = \frac{4}{3}$):
$\beta' = \frac{1.2}{4/3} = 0.9\,\text{mm}.$

2 What changes β?

- $\beta \propto \lambda$, $\beta \propto D$, $\beta \propto \frac{1}{d}$; red light gives WIDER fringes than blue.
- Slit WIDTH does not change β, only brightness; unequal slits: dark fringes not fully dark.
- Moving the screen changes β but not the angular width $\frac{\lambda}{d}$.

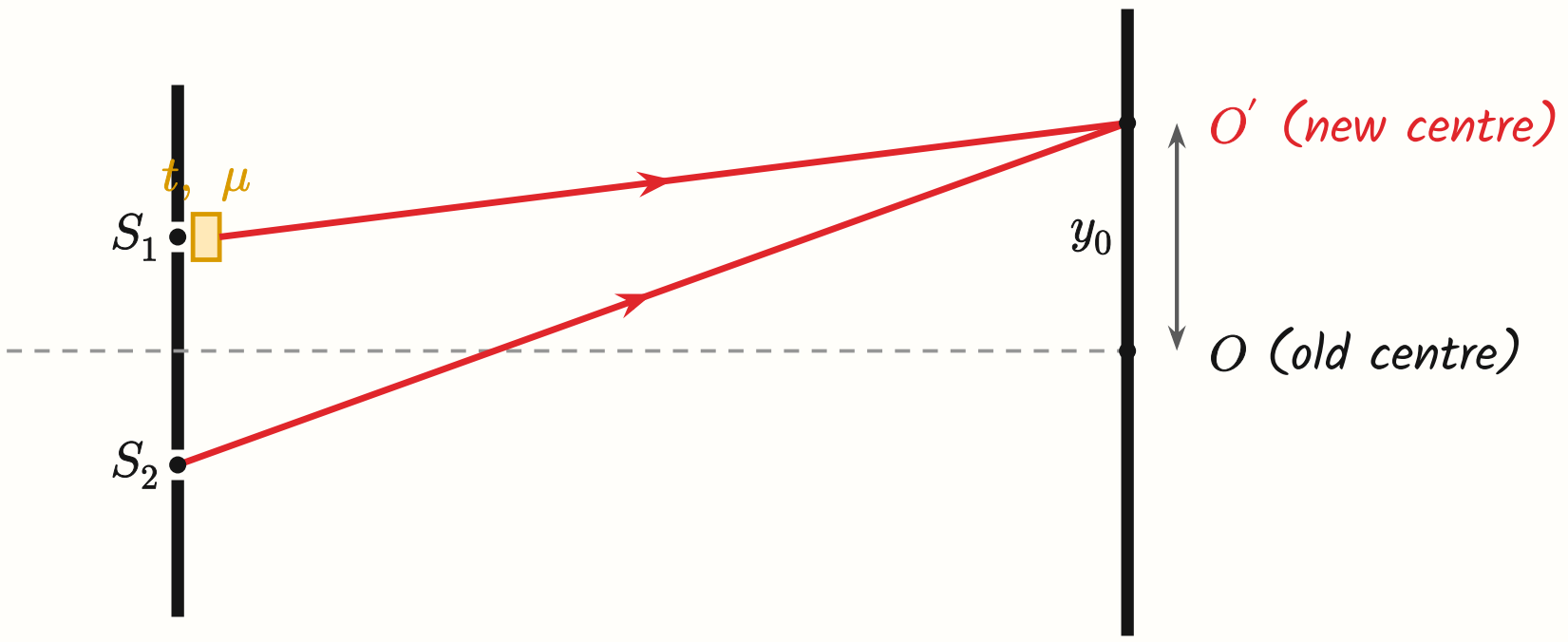
3 White light

- Centre ($\Delta x = 0$ for every λ): WHITE central fringe. On each side, coloured fringes: VIOLET edge nearer the centre (small β), RED farther out.
- After a few fringes the colours overlap: uniform white.
- With $d = 0.50\,\text{mm}$, $D = 1.0\,\text{m}$: first bright of violet ($400\,\text{nm}$) at $0.8\,\text{mm}$, of red ($700\,\text{nm}$) at $1.4\,\text{mm}$.

YDSE: Thin Sheet

YDSE · thin transparent sheet (thickness t, index μ) over S_1 · light normal to the sheet · absorption neglected

1 The pattern shifts towards the covered slit



At O' the extra optical path $(\mu-1)t$ of the sheet equals $S_2O' - S_1O'$.

2 Shift of the pattern

$$\Delta x = \frac{yd}{D} - (\mu-1)t \;\Rightarrow\; y_0 = \frac{(\mu-1)tD}{d}$$

$$N = \frac{y_0}{\beta} = \frac{(\mu-1)t}{\lambda}$$

y_0 = shift of the WHOLE pattern, towards the covered slit · N = fringes shifted · β UNCHANGED · $D \gg d$, small angles

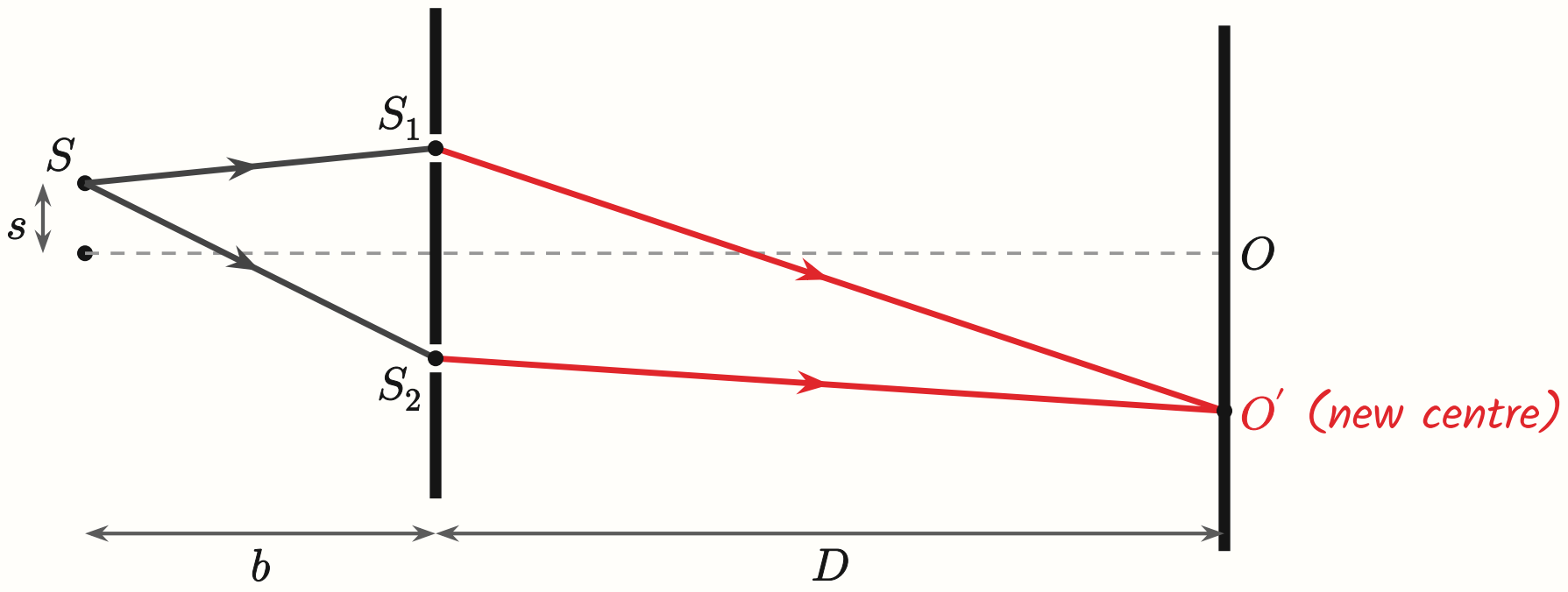
Example

$\mu = 1.5,\ t = 6.0\,\mu\text{m},\ \lambda = 600\,\text{nm},\ d = 0.50\,\text{mm},\ D = 1.0\,\text{m}$:
$N = \frac{0.5\times6.0\times10^{-6}}{600\times10^{-9}} = 5$ *fringes*, $y_0 = 6.0\,\text{mm}$.

YDSE: Source Slit Displaced

YDSE · $D \gg d$ · small angles · source shift $s \ll b$

1 Source moved by s (here upwards)



2 Shift of the central fringe

$$\Delta x_{total} = \frac{sd}{b} + \frac{yd}{D} = 0 \;\Rightarrow\; y_0 = -\frac{sD}{b}$$

S nearer S_1: $SS_2 - SS_1 = \frac{sd}{b}$ · b = source-to-slits distance · minus: OPPOSITE side · β unchanged

Example

$s = 1.0\,\text{mm}$ *up*, $b = 20\,\text{cm}$, $D = 1.0\,\text{m}$: *central fringe moves* $\frac{1.0\times1.0}{0.20} = 5.0\,\text{mm}$ *DOWN.*

3 Oblique plane wave

$$\Delta x = d\sin\theta - d\sin\alpha$$

plane wave at angle α to the axis · central maximum at $\theta = \alpha$ (along the incident direction)

Phase Change and Lloyd's Mirror

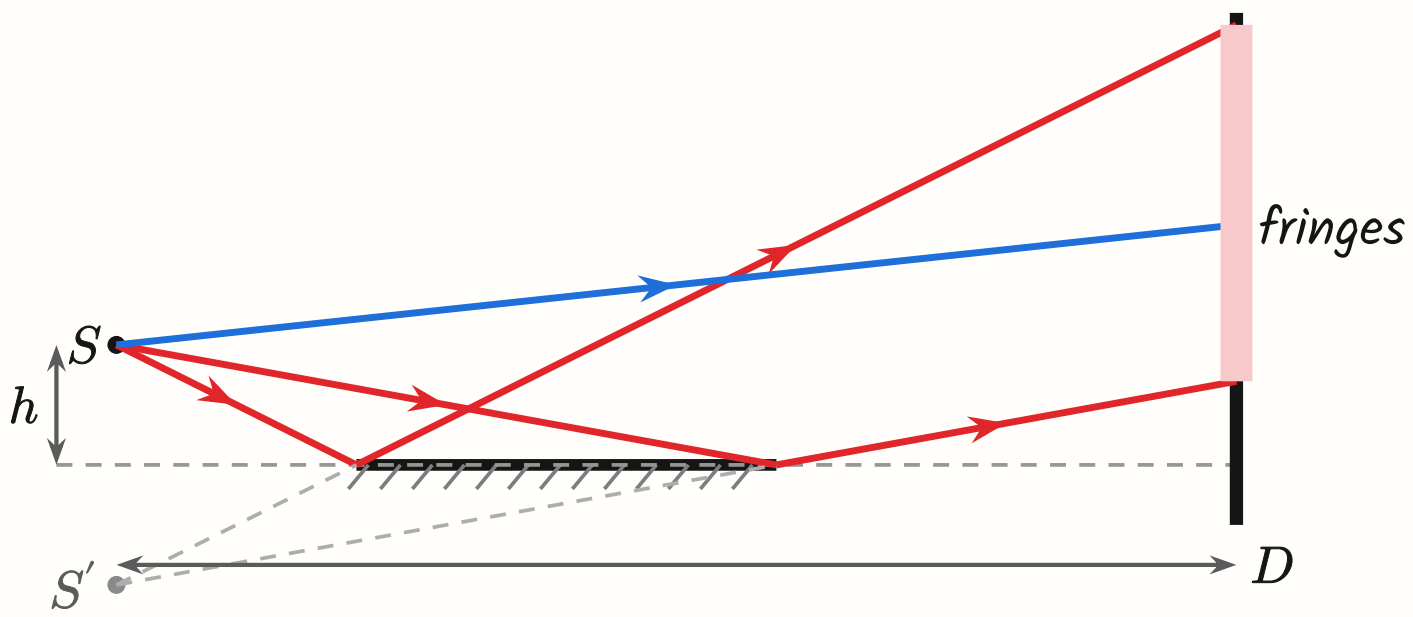
reflection at the boundary of two transparent media · monochromatic, coherent

1 Phase change on reflection

$$\text{rarer} \to \text{DENSER}: \quad \phi = \pi \quad (\text{extra } \tfrac{\lambda}{2})$$

from a medium of HIGHER μ (or a metal mirror) · denser → rarer: none · transmitted wave: none

2 Lloyd's mirror



Direct (blue) and reflected (red) light from ONE source S; the reflected light seems to come from S'.

$$d = 2h \qquad \beta = \frac{\lambda D}{2h}$$

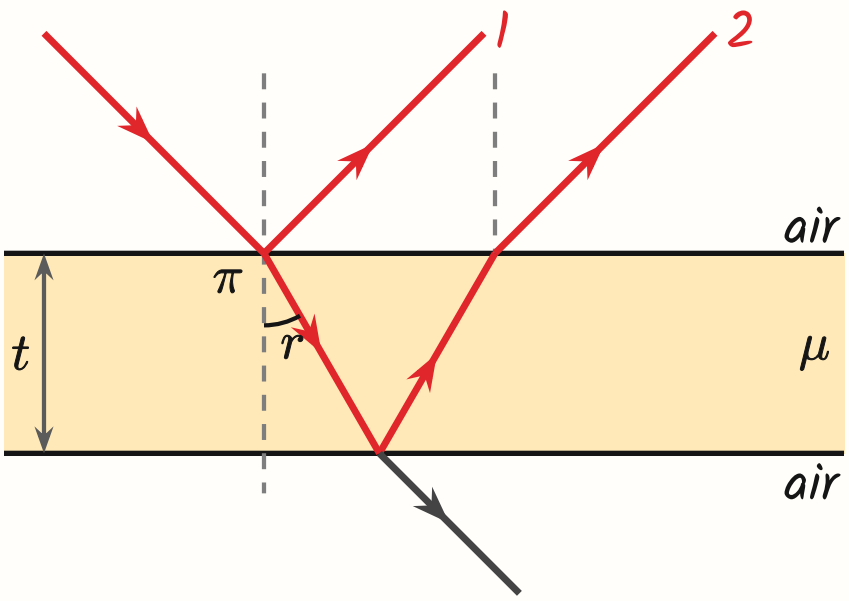
h = height of S above the mirror plane · $D \gg h$, small angles · bright where $\Delta x = (2n-1)\frac{\lambda}{2}$, dark where $\Delta x = n\lambda$ (opposite of YDSE)

- *A screen touching the end of the mirror is DARK at the contact point ($\Delta x = 0$): proof of the π phase change. Example: $h = 0.50\,\text{mm}$, $D = 1.0\,\text{m}$, $\lambda = 500\,\text{nm}$: $\beta = 0.50\,\text{mm}$.*

Thin Film Interference

film of UNIFORM thickness t, index μ, in AIR · r = angle of refraction inside the film · monochromatic

1 Two reflected rays from one ray



- Ray 1: reflected at the TOP (air → film, denser): phase change π.
- Ray 2: reflected at the BOTTOM (film → air): no phase change; extra optical path $2\mu t\cos r$.

2 Conditions (film in air)

$$\text{reflected bright: } 2\mu t\cos r = \left(n-\tfrac{1}{2}\right)\lambda$$

$$\text{reflected dark: } 2\mu t\cos r = n\lambda$$

bright: $n = 1, 2, 3, \ldots$ · dark: $n = 0, 1, 2, \ldots$ · normal incidence: $\cos r = 1$, path $2\mu t$ · λ in air

- TRANSMITTED light: the conditions are SWAPPED (bright at $2\mu t\cos r = n\lambda$): energy not reflected is transmitted.
- A film on glass with $\mu_{air} < \mu < \mu_{glass}$: BOTH reflections have π, so the reflected conditions swap too.

Golden points

★ Film in air, normal incidence: THINNEST film that reflects λ strongly has $2\mu t = \frac{\lambda}{2}$, i.e. $t = \frac{\lambda}{4\mu}$.

Thin Films: Coatings and Colours

thin films of uniform thickness · NORMAL incidence · λ in air

1 Anti-reflection coating

$$2\mu_f t = \left(n - \tfrac{1}{2}\right)\lambda \;\Rightarrow\; t_{min} = \frac{\lambda}{4\mu_f}$$

coating index μ_f BETWEEN air and glass: both reflections have π, they cancel for this λ · $n = 1, 2, \ldots$ · complete cancellation needs $\mu_f = \sqrt{\mu_{glass}}$

Example

Coating $\mu_f = 1.38$ on glass ($\mu = 1.5$), $\lambda = 550\,\text{nm}$:

$t_{min} = \frac{550}{4\times 1.38} = 99.6\,\text{nm}$.

2 Colours of a soap film

Example

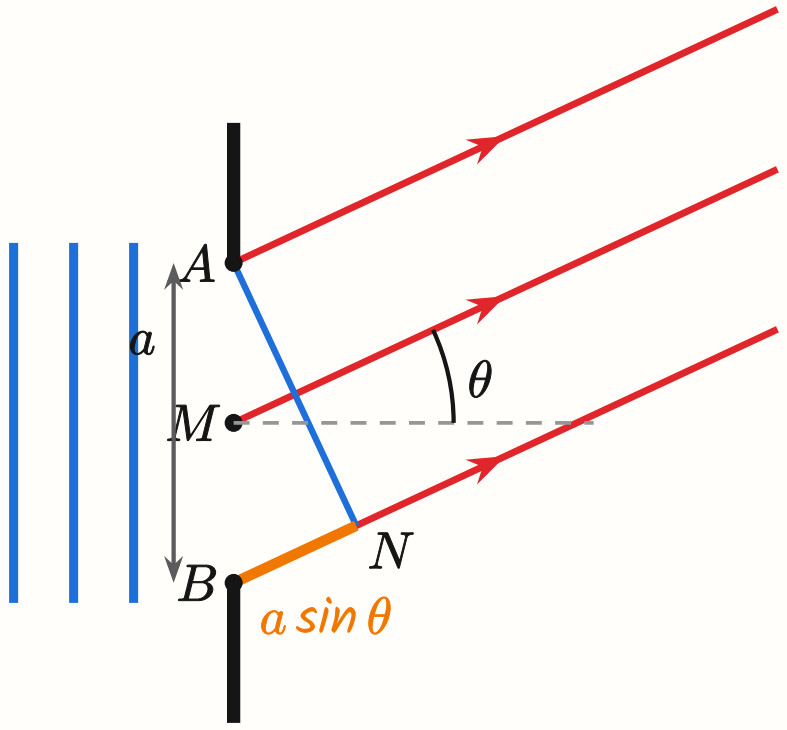
Soap film in air, $\mu = 1.33$, $t = 300\,\text{nm}$, white light at normal incidence: reflected bright for $\lambda = \frac{2\mu t}{n-\frac{1}{2}} = \frac{798}{n-0.5}\,\text{nm}$: 1596, 532, 319 nm. Only 532 nm is visible: the film looks GREEN.

- *Colour depends on the thickness t and on the viewing angle ($\cos r$): oil on water shows changing colours.*
- *Very thin film ($t \ll \lambda$) in air: path difference ≈ 0, only the π change: DARK in reflection (black spot of a soap film just before it bursts).*
- *Thick film ($t \gg \lambda$, white light): maxima of many colours overlap, no colours seen.*

Diffraction at a Single Slit

single slit of width a · plane monochromatic wave at NORMAL incidence · Fraunhofer (far-field or lens)

1 Path difference across the slit



- All parallel rays at angle θ meet at one point P of a distant screen.
- Edge-to-edge path difference $BN = a\sin\theta$.
- $\theta = 0$: all wavelets in phase: CENTRAL MAXIMUM.

2 Minima

$$a\sin\theta = n\lambda \qquad n = \pm 1, \pm 2, \ldots \quad (n \neq 0)$$

first minimum at $a\sin\theta = \lambda$: pair each point of the upper half AM with the point $\frac{a}{2}$ below it; path difference $\frac{\lambda}{2}$, every pair cancels · n-th minimum: divide into $2n$ parts

secondary maxima: $a\sin\theta \approx \left(n + \frac{1}{2}\right)\lambda$

$n = 1, 2, \ldots$ · *APPROXIMATE positions (exact: 1.43λ, 2.46λ) · the middle third of the slit is left uncancelled, so these are much weaker*

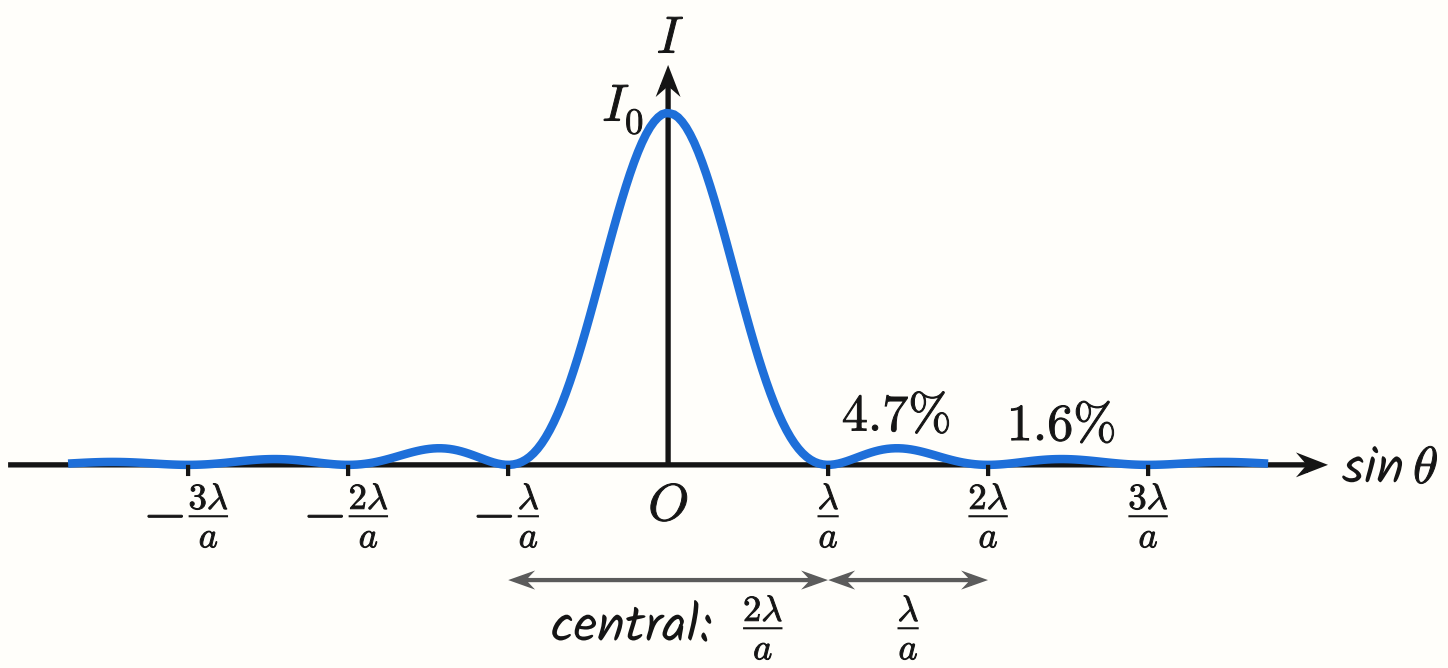
Example

$a = 0.20\,\text{mm}$, $\lambda = 500\,\text{nm}$, *screen at* $D = 2.0\,\text{m}$: *first minimum at* $\theta_1 = \frac{\lambda}{a} = 2.5 \times 10^{-3}\,\text{rad}$, *i.e.* $5.0\,\text{mm}$ *from the centre; central maximum* $10\,\text{mm}$ *wide.*

Single Slit: Intensity

single slit, width a · normal incidence · Fraunhofer · SMALL angles for the widths

1 Intensity pattern



CENTRAL maximum TWICE as wide as each secondary one; secondary peaks only 4.7%, 1.6% *of* I_0.

---

2 Formula and widths

$$I = I_0 \left(\frac{\sin\beta}{\beta}\right)^2 \qquad \beta = \frac{\pi a \sin\theta}{\lambda}$$

β = HALF the phase difference between the edge wavelets (rad) · I_0 = intensity at the centre

$$2\theta_1 = \frac{2\lambda}{a} \qquad w = \frac{2\lambda D}{a}$$

θ_1 = angle of the first minimum ($\sin\theta \approx \theta$) · w = linear width of the central maximum on a screen at distance $D \gg a$ (with a lens: $D \to f$) · other maxima: width $\frac{\lambda D}{a}$

Fresnel Distance; Diffraction vs Interference

aperture (slit) width a · monochromatic · screen at distance D

1 Fresnel distance

$$z_F = \frac{a^2}{\lambda}$$

$D \ll z_F$: spreading $\frac{\lambda D}{a} < a$, RAY OPTICS holds · $D \gg z_F$: diffraction dominates (Fraunhofer)

Example

$a = 3.0\,\text{mm},\ \lambda = 500\,\text{nm}$: $z_F = \frac{(3.0\times10^{-3})^2}{500\times10^{-9}} = 18\,\text{m}.$

2 Diffraction vs interference

	Interference (YDSE)	Single-slit diffraction
Waves	from TWO coherent sources	from points of ONE wavefront
Widths	all fringes equal, $\frac{\lambda D}{d}$	central $\frac{2\lambda D}{a}$, others $\frac{\lambda D}{a}$
Brightness	equal maxima (small angles)	falls fast away from the centre
Dark at	$\Delta x = (2n-1)\frac{\lambda}{2}$	$a\sin\theta = n\lambda,\ n \neq 0$

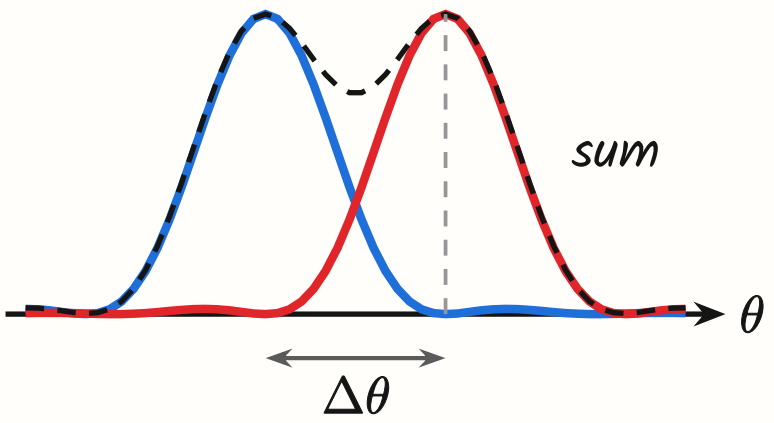
3 Double slit of finite width

- Real YDSE: interference fringes (spacing $\frac{\lambda D}{d}$) inside the single-slit envelope (central width $\frac{2\lambda D}{a}$), $d > a$.
- Bright fringes inside the central envelope: orders $|n| < \frac{d}{a}$. If $\frac{d}{a}$ is an integer, the orders $\pm\frac{d}{a}$ fall on diffraction minima and are MISSING.

Resolving Power

circular aperture of diameter D_o · INCOHERENT point sources · small angles

1 Rayleigh criterion



- JUST resolved: the central maximum of one image falls on the FIRST MINIMUM of the other.
- Sum (dashed) dips to 73.5%.

2 Telescope

$$\Delta\theta = \frac{1.22\lambda}{D_o} \qquad RP = \frac{1}{\Delta\theta} = \frac{D_o}{1.22\lambda}$$

$\Delta\theta$ = smallest resolvable angular separation of two stars (rad) · D_o = objective diameter

3 Microscope

$$d_{min} = \frac{1.22\lambda}{2\mu \sin\beta} \qquad RP = \frac{2\mu \sin\beta}{1.22\lambda}$$

d_{min} = smallest resolvable separation of two object points · β = half-angle of the cone of light into the objective · $\mu \sin\beta$ = numerical aperture

Example

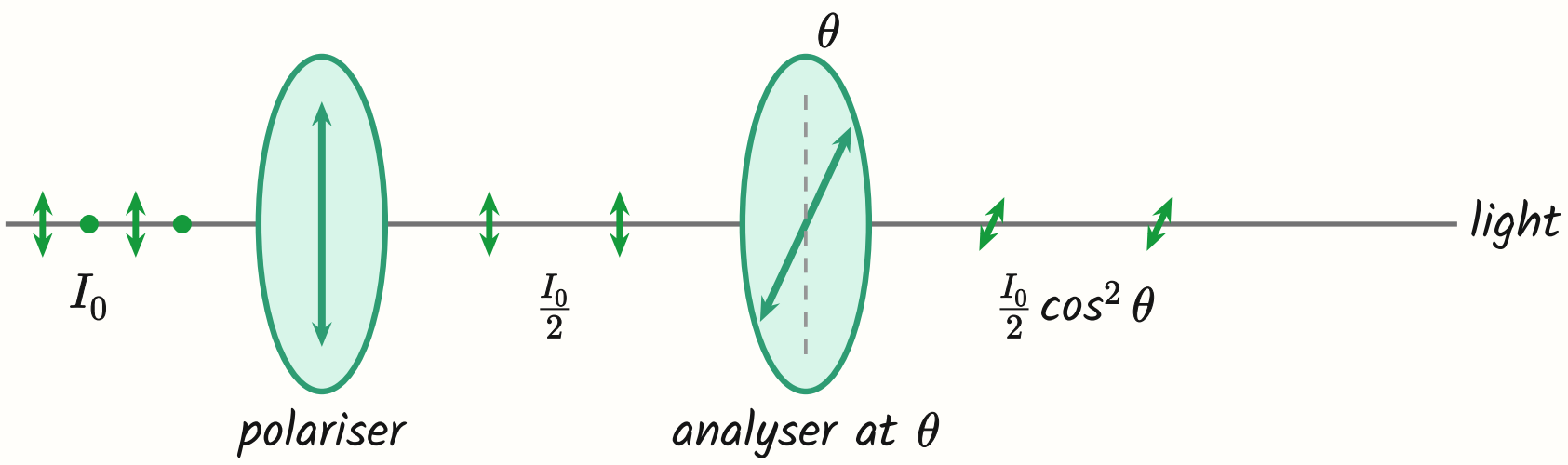
Telescope, $D_o = 0.10\,\text{m}$, $\lambda = 550\,\text{nm}$: $\Delta\theta = 6.71 \times 10^{-6}\,\text{rad}$.

Microscope, $\lambda = 500\,\text{nm}$, oil $\mu = 1.5$, $\beta = 60°$: $d_{min} = 235\,\text{nm}$.

Polarisation and Malus's Law

IDEAL polaroids · UNPOLARISED light of intensity I_0 enters · θ = angle between transmission axes

1 Unpolarised light through two polaroids



Arrows: $\vec{E}$ in the page; dots: $\vec{E}$ perpendicular to the page.

2 Malus's law

$$I = I_1 \cos^2\theta$$

I_1 = intensity of PLANE-polarised light falling on the analyser · θ = angle between its $\vec{E}$ and the transmission axis · unpolarised light through ONE ideal polaroid: $\frac{I_0}{2}$ (average of $\cos^2$)

- Only the component $E\cos\theta$ along the axis passes, and $I \propto E^2$.
- Crossed polaroids ($\theta = 90°$): no light. $\theta = 60°$: $\frac{I_0}{2} \times \frac{1}{4} = \frac{I_0}{8}$.
- Polarisation proves that light is TRANSVERSE (sound in air cannot be polarised).

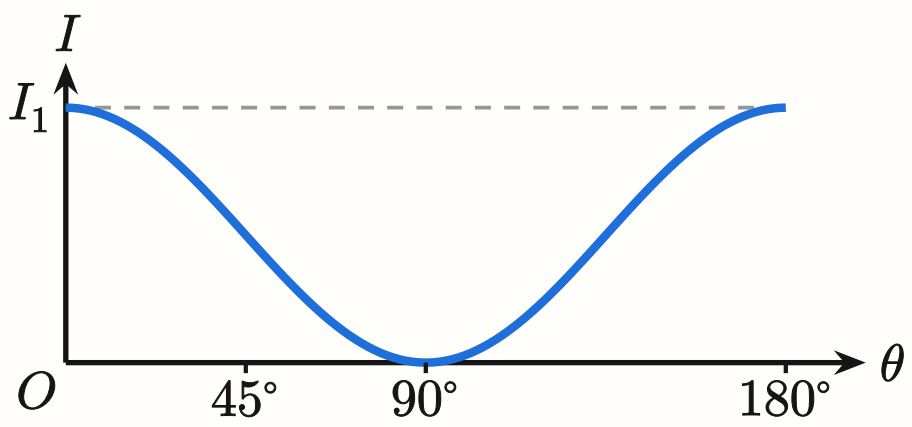
Golden points

★ Polaroid: aligned long-chain molecules absorb $\vec{E}$ ALONG the chains; transmission axis $\perp$ chains.

Malus's Law: Results

IDEAL polaroids · θ = angle between transmission axes · I_0 = UNPOLARISED intensity entering

1 Graph of Malus's law



- Plane-polarised (I_1), ROTATING analyser: maximum twice, ZERO twice per turn.
- Unpolarised: CONSTANT $\frac{I_0}{2}$. Partially polarised: varies, never zero.

2 Three polaroids

$$I = \frac{I_0}{2}\cos^2\theta\,\sin^2\theta = \frac{I_0}{8}\sin^2 2\theta$$

1st and 3rd CROSSED, middle at θ to the 1st · maximum $\frac{I_0}{8}$ at $\theta = 45°$ · no middle one: zero · middle at 30°: $\frac{I_0}{2} \times \frac{3}{4} \times \frac{1}{4} = \frac{3I_0}{32} = 0.094I_0$

3 Degree of polarisation

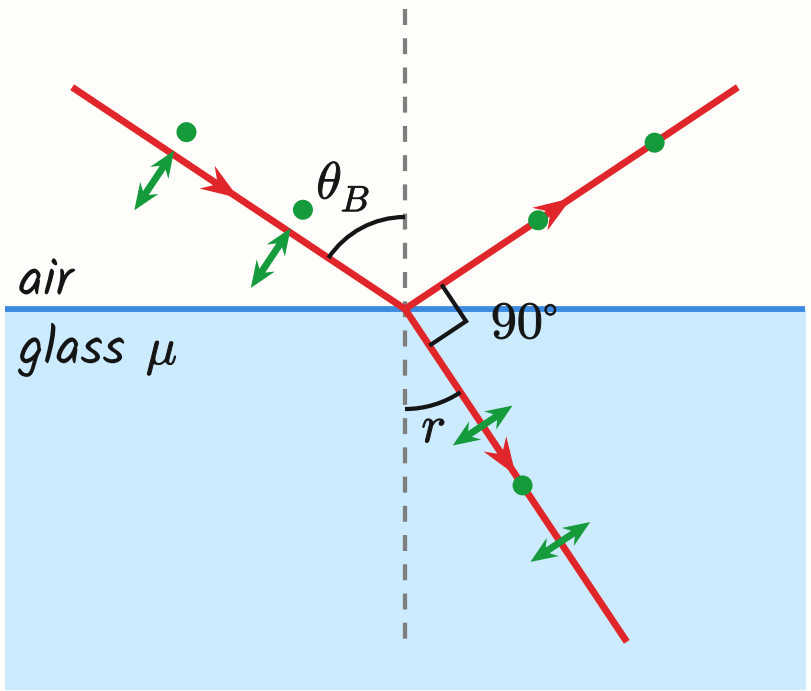
$$P = \frac{I_{max} - I_{min}}{I_{max} + I_{min}}$$

I_{max}, I_{min} seen through a rotating analyser · $P = 1$: plane polarised, $P = 0$: unpolarised

Brewster's Law

UNPOLARISED light on a transparent surface · from medium μ_1 to μ_2 · air to glass in the figure

1 Polarisation by reflection



- At θ_B the REFLECTED and REFRACTED rays are at $90°$.
- Reflected light: COMPLETELY plane polarised, $\vec{E}$ perpendicular to the plane of incidence (dots).
- Refracted light: PARTIALLY polarised.

2 Brewster's law

$$\tan\theta_B = \frac{\mu_2}{\mu_1} \qquad \text{air to medium: } \tan\theta_B = \mu$$

θ_B = Brewster (polarising) angle, measured from the normal · proof: $\mu = \frac{\sin\theta_B}{\sin r}$ and $r = 90° - \theta_B$, so $\mu = \tan\theta_B$

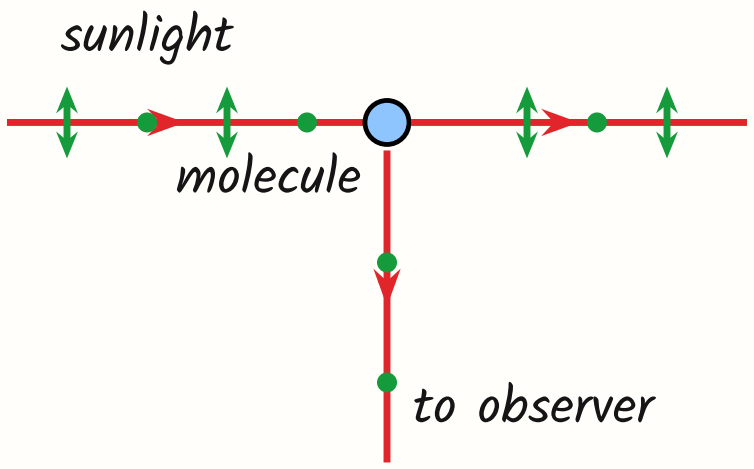
Example

Glass $\mu = 1.5$: $\theta_B = \tan^{-1} 1.5 = 56.3°$, $r = 33.7°$. Water $\mu = 1.33$: $\theta_B = 53.1°$. Glass to air (inside the glass): $\tan\theta_B = \frac{1}{1.5}$, $\theta_B = 33.7°$.

Polarisation by Scattering

UNPOLARISED sunlight · scattering by molecules much SMALLER than λ · observer at $90°$

1 Light scattered at $90°$ is plane polarised



- Electrons oscillate in the plane perpendicular to the beam.
- No radiation along the line of oscillation: light going down carries only the dots ($\vec{E} \perp$ page).

2 Blue sky

$$I_{scattered} \propto \frac{1}{\lambda^4}$$

Rayleigh scattering by particles much SMALLER than λ · blue ($400\,\text{nm}$) is scattered $\left(\frac{700}{400}\right)^4 = 9.4$ times more than red ($700\,\text{nm}$)

- The sky looks BLUE; the sun near the horizon looks RED (blue scattered out of the long path).
- Large particles (cloud droplets, size $\gg \lambda$) scatter all colours equally: clouds are WHITE.

Golden points

★ Uses of polaroids: glare-free sunglasses, camera filters, 3D films, liquid-crystal displays.

Doppler Effect for Light

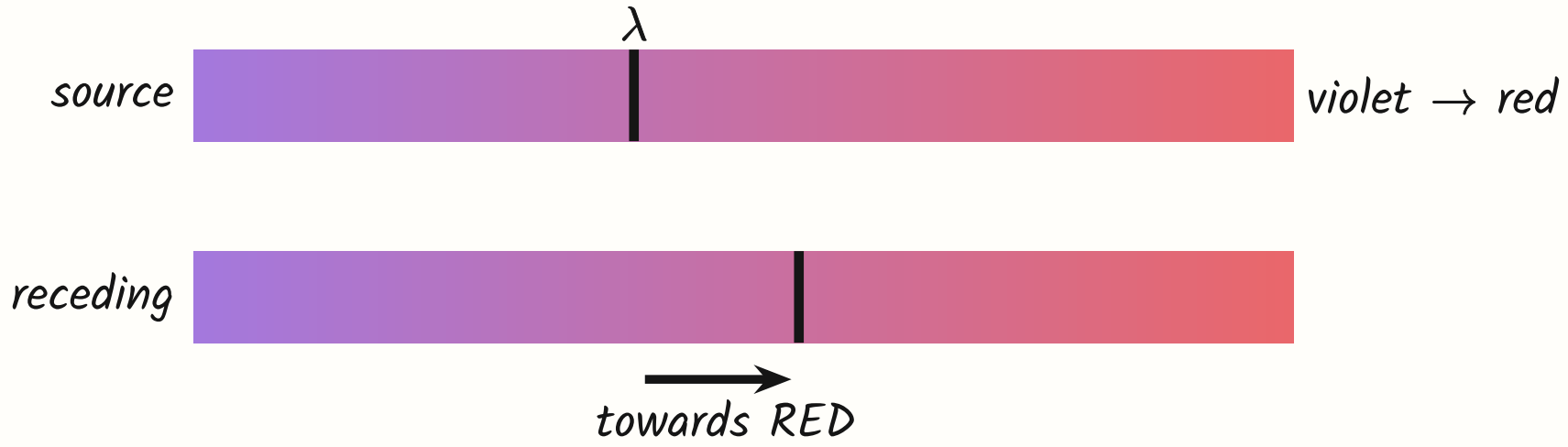
source and observer moving apart or together · $v_r \ll c$ · v_r = *RADIAL relative speed,* $v_r > 0$ *when RECEDING*

1 Shift of frequency and wavelength

$$\frac{\Delta\nu}{\nu} = -\frac{v_r}{c} \qquad \frac{\Delta\lambda}{\lambda} = +\frac{v_r}{c}$$

$\Delta\nu = \nu_{observed} - \nu_{source}$, $\Delta\lambda = \lambda_{observed} - \lambda_{source}$ · *only the RELATIVE velocity matters (no medium needed)* · *motion across the line of sight: no first-order shift*

2 Red shift and blue shift



- *RECEDING source* ($v_r > 0$): λ *increases, RED SHIFT (distant galaxies: expanding universe).*
- *APPROACHING source* ($v_r < 0$): λ *decreases, BLUE SHIFT.*

Example

A hydrogen line of $656.3\,\mathrm{nm}$ *from a star is observed at* $658.5\,\mathrm{nm}$*: red shift,* $v_r = c\frac{\Delta\lambda}{\lambda} = 3\times10^8 \times \frac{2.2}{656.3} = 1.01\times10^6\,\mathrm{m\,s^{-1}}$ *AWAY from us.*

Special Cases

YDSE unless stated · $D \gg d$ · small angles · conditions in each case

1 Changes to the slits

- One slit CLOSED: no interference; only the single-slit diffraction pattern of the open slit (central intensity I_0 instead of $4I_0$).
- Sources with an extra phase difference ϕ_0 (e.g. π): the pattern shifts by $\frac{\phi_0}{2\pi}\beta$; for $\phi_0 = \pi$ the centre is DARK.
- Identical sheets on BOTH slits: no shift.

2 Two wavelengths together

$$n_1\lambda_1 = n_2\lambda_2 \text{ (bright fringes coincide)}$$

nearest coincidence to the centre: smallest integers n_1, n_2 · same d, D for both

Example

$\lambda_1 = 500\,\text{nm}$, $\lambda_2 = 600\,\text{nm}$, $d = 1.0\,\text{mm}$, $D = 1.0\,\text{m}$:
$6 \times 500 = 5 \times 600$, first coincidence at $y = \frac{6\times500\times10^{-9}\times1.0}{1.0\times10^{-3}} = 3.0\,\text{mm}$.

3 Missing wavelengths and fringe shapes

- White light: at a point y, the wavelengths with $\frac{yd}{D} = (2n-1)\frac{\lambda}{2}$ are MISSING (dark there).
- Fringes on a screen parallel to S_1S_2: hyperbolas (nearly straight near the centre). Screen perpendicular to S_1S_2: CIRCULAR fringes.

Standard Results

1 One-line results

Quantity	Result	Condition
Phase, path	$\phi = \frac{2\pi}{\lambda}\Delta x$	Δx = optical path (μL)
Snell (Huygens)	$\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{\mu_2}{\mu_1}$	plane wave, plane boundary
Intensity	$I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$	coherent, same polarisation
Max, min	$(\sqrt{I_1} \pm \sqrt{I_2})^2$	coherent
YDSE path	$d\sin\theta \approx \frac{yd}{D}$	$D \gg d$, small θ
Fringe width	$\beta = \frac{\lambda D}{d}$, $\theta_\beta = \frac{\lambda}{d}$	small angles
In a medium	β/μ	whole set-up immersed
Thin sheet	$y_0 = \frac{(\mu-1)tD}{d}$	towards the covered slit
Source shift	$y_0 = -\frac{sD}{b}$	opposite side
Thin film (air)	$2\mu t\cos r = (n - \frac{1}{2})\lambda$	reflected bright
Single slit	$a\sin\theta = n\lambda$	minima, $n \neq 0$
Central max	$\frac{2\lambda D}{a}$	small angles
Fresnel distance	$\frac{a^2}{\lambda}$	ray optics if $D \ll z_F$
Telescope	$\Delta\theta = \frac{1.22\lambda}{D_o}$	Rayleigh, circular
Microscope	$d_{min} = \frac{1.22\lambda}{2\mu\sin\beta}$	Rayleigh
Malus	$I = I_1\cos^2\theta$	ideal analyser
Brewster	$\tan\theta_B = \mu$	from air; refl. $\perp$ refr.
Doppler	$\frac{\Delta\lambda}{\lambda} = \frac{v_r}{c}$	$v_r \ll c$, receding $+$

Summary

$\beta = \frac{\lambda D}{d}$ • $a\sin\theta = n\lambda$ • $I = I_1\cos^2\theta$ • $\tan\theta_B = \mu$

coherent, monochromatic · $D \gg d$, small angles · Fraunhofer · ideal polaroids

Common Traps

1 Exam traps

Avoid these

- ✗ Single-slit MINIMA $a\sin\theta = n\lambda$ look like YDSE MAXIMA $d\sin\theta = n\lambda$ (opposite meaning).
- ✗ Central maximum of a single slit is $\frac{\lambda D}{a}$ wide (no: $\frac{2\lambda D}{a}$).
- ✗ Immersing YDSE in water widens the fringes (no: β/μ, narrower).
- ✗ A sheet shifts fringes away from the covered slit (no: towards it); a source shift moves them to the OPPOSITE side.
- ✗ Slit width changes β (no: only brightness and contrast).
- ✗ Forgetting the extra $\frac{\lambda}{2}$ at reflection from a DENSER medium in thin films.
- ✗ Two separate bulbs give fringes (no: incoherent).
- ✗ Unpolarised light through a polaroid gives $I_0\cos^2\theta$ (no: $\frac{I_0}{2}$).
- ✗ At Brewster's angle the reflected ray is perpendicular to the INCIDENT ray (no: to the REFRACTED ray).
- ✗ Using $y = \frac{n\lambda D}{d}$ at large angles (no: $d\sin\theta = n\lambda$, $y = D\tan\theta$).
- ✗ Frequency changes on refraction (no: ν fixed, $\lambda \to \frac{\lambda}{\mu}$).
- ✗ Red shift means approaching (no: receding).

Golden points

- ★ First ask: coherent? which slit is covered? reflection from a denser medium? small angles? polarised or unpolarised?

Questions: JEE Main Level (I)

monochromatic, coherent · in air · $D \gg d$, small angles

Q1 · JEE Main level · MCQ

YDSE: $\lambda = 500\,\text{nm}$, $d = 1.0\,\text{mm}$, $D = 2.0\,\text{m}$. Distance from the 2nd bright fringe on one side to the 3rd dark fringe on the OTHER side: (A) 4.5 (B) 0.5 (C) 2.5 (D) $3.5\,\text{mm}$

Solution: $\beta = 1.0\,\text{mm}$; $2\beta + 2.5\beta = 4.5\,\text{mm}$.

Answer: (A)

Q2 · JEE Main level · Numerical

Two coherent sources have intensities in the ratio $9:1$. Find $\frac{I_{max}}{I_{min}}$ in their interference pattern.

Solution: Amplitude ratio $3:1$: $\frac{(3+1)^2}{(3-1)^2} = \frac{16}{4} = 4$.

Answer: 4

Q3 · JEE Main level · MCQ

The fringe width in a YDSE in air is $2.0\,\text{mm}$. The whole apparatus is immersed in a liquid of refractive index 1.25. New fringe width: (A) $2.5\,\text{mm}$ (B) $2.0\,\text{mm}$ (C) $1.6\,\text{mm}$ (D) $0.8\,\text{mm}$

Solution: $\beta' = \frac{\beta}{\mu} = \frac{2.0}{1.25} = 1.6\,\text{mm}$.

Answer: (C)

Q4 · JEE Main level · Numerical

A thin sheet ($\mu = 1.6$) over one slit of a YDSE ($\lambda = 540\,\text{nm}$) moves the central bright fringe to the old 4th bright fringe. Thickness in μm?

Solution: $(\mu - 1)t = 4\lambda$: $t = \frac{4 \times 540 \times 10^{-9}}{0.6} = 3.6 \times 10^{-6}\,\text{m}$.

Answer: 3.6

Questions: JEE Main Level (2)

monochromatic · in air · small angles · ideal polaroids · $c = 3 \times 10^8\,\mathrm{m\,s^{-1}}$

Q5 · JEE Main level · MCQ

Light of $600\,\mathrm{nm}$ *falls normally on a slit* $0.12\,\mathrm{mm}$ *wide. Central maximum width on a screen* $1.5\,\mathrm{m}$ *away: (A)* 7.5 *(B)* 15 *(C)* 30 *(D)* $1.5\,\mathrm{mm}$

Solution: $w = \frac{2\lambda D}{a} = \frac{2\times600\times10^{-9}\times1.5}{0.12\times10^{-3}} = 15\,\mathrm{mm}.$

Answer: (B)

Q6 · JEE Main level · Numerical

Unpolarised light in air falls on glass of refractive index $\sqrt{3}$*. Find the Brewster angle and the angle of refraction then.*

Solution: $\tan\theta_B = \sqrt{3}$*:* $\theta_B = 60^\circ$*;* $r = 90^\circ - 60^\circ = 30^\circ.$

Answer: $60^\circ,\ 30^\circ$

Q7 · JEE Main level · MCQ

Unpolarised light $(64\,\mathrm{W\,m^{-2}})$ *passes two ideal polaroids with axes at* 30°*. Output: (A)* 48 *(B)* 32 *(C)* 24 *(D)* $16\,\mathrm{W\,m^{-2}}$

Solution: $64 \times \frac{1}{2} \times \cos^2 30^\circ = 32 \times \frac{3}{4} = 24\,\mathrm{W\,m^{-2}}.$

Answer: (C)

Q8 · JEE Main level · Numerical

A line of $589.0\,\mathrm{nm}$ *(lab) is observed at* $588.2\,\mathrm{nm}$ *from a star* $(v \ll c)$*. Find its radial speed and direction.*

Solution: $\frac{\Delta\lambda}{\lambda} = \frac{-0.8}{589.0}$*: blue shift, approaching;*

$v_r = 3 \times 10^8 \times \frac{0.8}{589.0} = 4.07 \times 10^5\,\mathrm{m\,s^{-1}}.$

Answer: $4.07 \times 10^5\,\mathrm{m\,s^{-1}}$**, towards us**

Questions: JEE Advanced Level (I)

monochromatic, coherent · $D \gg d$, small angles · thin sheets normal to the light, no absorption

A1 · JEE Advanced level · Numerical, multi-step

YDSE in air: $\lambda = 560\,\text{nm}$, $d = 1.0\,\text{mm}$, $D = 1.0\,\text{m}$. A sheet of index 1.4 over the upper slit moves the central bright fringe to the old 5th bright fringe. Find (i) the thickness t, (ii) the fringe width and the shift of the central fringe when the WHOLE set-up (sheet included) is put in water ($\mu_w = \frac{4}{3}$), (iii) that shift in fringes.

Solution: (i) $(1.4-1)t = 5\lambda$: $t = \frac{5\times560}{0.4}\,\text{nm} = 7.0\,\mu\text{m}$ (shift in air $5\beta = 2.8\,\text{mm}$). (ii) $\beta' = \frac{\lambda D}{\mu_w d} = 0.42\,\text{mm}$. Optical paths: $\mu_w(S_2P - S_1P) = (1.4-\mu_w)t$, so $y_0' = \frac{(1.4-\mu_w)tD}{\mu_w d} = \frac{0.0667\times7.0\times10^{-6}}{1.333\times10^{-3}} = 0.35\,\text{mm}$. (iii) $\frac{0.35}{0.42} = 0.83$ fringe.

Answer: (i) $7.0\,\mu\text{m}$ (ii) $0.42\,\text{mm}$, $0.35\,\text{mm}$ up (iii) 0.83

A2 · JEE Advanced level · Numerical, multi-step

YDSE in air: $\lambda = 450\,\text{nm}$, $d = 0.90\,\text{mm}$, $D = 1.2\,\text{m}$; slit 1 alone gives I_0, slit 2 alone $2I_0$ at the screen. Find (i) β, (ii) I_{max}, I_{min} and the visibility, (iii) the intensity at $y = 0.20\,\text{mm}$.

Solution: (i) $\beta = \frac{450\times10^{-9}\times1.2}{0.90\times10^{-3}} = 0.60\,\text{mm}$. (ii) $I_{max} = (1+\sqrt{2})^2 I_0 = 5.83 I_0$, $I_{min} = (\sqrt{2}-1)^2 I_0 = 0.172 I_0$, $V = \frac{2\sqrt{2}}{3} = 0.943$. (iii) $\Delta x = \frac{yd}{D} = 150\,\text{nm} = \frac{\lambda}{3}$, $\phi = 120°$: $I = 3I_0 + 2\sqrt{2}I_0\cos 120° = 1.59 I_0$.

Answer: (i) $0.60\,\text{mm}$ (ii) $5.83I_0$, $0.172I_0$, 0.943 (iii) $1.59I_0$

Questions: JEE Advanced Level (2)

thin film of uniform thickness · normal incidence · Fraunhofer · small angles

A3 · JEE Advanced level · Numerical, multi-step

An oil film ($\mu = 1.25$) of uniform thickness lies on glass ($\mu = 1.5$). In white light at normal incidence the REFLECTED light has minima at $700\,\text{nm}$ and $500\,\text{nm}$ and none between them. Find (i) the order, (ii) the thickness, (iii) the visible wavelengths (400–$700\,\text{nm}$) most strongly reflected.

Solution: Both reflections (air→oil, oil→glass) have a π change, so reflected minima: $2\mu t = (m + \frac{1}{2})\lambda$. (i) Consecutive: $(m+\frac{1}{2})700 = (m+\frac{3}{2})500$: $m = 2$. (ii) $2\mu t = 2.5 \times 700 = 1750\,\text{nm}$: $t = \frac{1750}{2\times 1.25} = 700\,\text{nm}$. (iii) Maxima: $2\mu t = m\lambda$: $\lambda = \frac{1750}{m} = 875$, 583.3, 437.5, $350\,\text{nm}$; visible: 583.3 and $437.5\,\text{nm}$.

Answer: (i) $m = 2$ (ii) $700\,\text{nm}$ (iii) $583\,\text{nm}$, $438\,\text{nm}$

A4 · JEE Advanced level · Numerical, multi-step

A double slit has slits of width $a = 0.10\,\text{mm}$ with centres $d = 0.50\,\text{mm}$ apart; $\lambda = 550\,\text{nm}$, $D = 1.0\,\text{m}$. Find (i) the fringe width, (ii) the width of the central diffraction maximum, (iii) the number of bright fringes inside it.

Solution: (i) $\beta = \frac{550\times10^{-9}\times1.0}{0.50\times10^{-3}} = 1.1\,\text{mm}$. (ii) $\frac{2\lambda D}{a} = \frac{2\times550\times10^{-9}\times1.0}{0.10\times10^{-3}} = 11\,\text{mm}$. (iii) $\frac{d}{a} = 5$: orders $|n| < 5$ inside; $n = \pm5$ fall on the diffraction minima (missing): $n = 0, \pm1, \ldots, \pm4$, 9 fringes.

Answer: (i) $1.1\,\text{mm}$ (ii) $11\,\text{mm}$ (iii) 9

Questions: JEE Advanced Level (3)

ideal polaroids · unpolarised light enters · μ values given in each question

AS · JEE Advanced level · Numerical, multi-step

Unpolarised light ($80\,\mathrm{W\,m^{-2}}$) passes through polaroid 1, then polaroid 2, which rotates at $\omega = 10\pi\,\mathrm{rad\,s^{-1}}$, then polaroid 3, CROSSED with polaroid 1. Find (i) the maximum, (ii) the time-averaged transmitted intensity, (iii) the number of maxima per second.

Solution:
$I = \frac{80}{2}\cos^2\omega t \sin^2\omega t = \frac{80}{8}\sin^2 2\omega t = 5(1-\cos 4\omega t)\,\mathrm{W\,m^{-2}}$. (i) $\frac{80}{8} = 10\,\mathrm{W\,m^{-2}}$. (ii) $5\,\mathrm{W\,m^{-2}}$. (iii) Maxima when $2\omega t = \frac{\pi}{2} + k\pi$: every $\frac{\pi}{2\omega} = 0.05\,\mathrm{s}$, i.e. 20 per second.

Answer: (i) $10\,\mathrm{W\,m^{-2}}$ (ii) $5\,\mathrm{W\,m^{-2}}$ (iii) 20

A6 · JEE Advanced level · Numerical, multi-step

Unpolarised light travelling in water ($\mu_1 = \frac{4}{3}$) falls on a glass block ($\mu_2 = 1.5$). Find (i) the Brewster angle, (ii) the angle of refraction then, (iii) the fraction of the REFLECTED intensity that passes an ideal polaroid whose axis is at 30° to the reflected $\vec{E}$.

Solution: (i) $\tan\theta_B = \frac{1.5}{4/3} = 1.125$: $\theta_B = 48.4^\circ$. (ii) $r = 90^\circ - 48.4^\circ = 41.6^\circ$ (check: $\frac{4}{3}\sin 48.4^\circ = 1.5\sin 41.6^\circ = 0.997$). (iii) Reflected light is plane polarised: Malus, $\cos^2 30^\circ = 0.75$.

Answer: (i) 48.4° (ii) 41.6° (iii) 0.75